

Markscheme

Mathematics: analysis and approaches
Practice question bank

151. (a)

Given $z_1 = 2e^{i\pi/3}$ and $z_2 = 5e^{i\pi/6}$.

Find $z_1 z_2$ in the form $a + bi$.

$$z_1 z_2 = (2 \times 5)e^{i(\frac{\pi}{3} + \frac{\pi}{6})} \quad (M1)$$

$$= 10e^{i\pi/2} \quad \mathbf{A1}$$

$$= 10(\cos 90^\circ + i \sin 90^\circ) \quad (M1)$$

$$= 10i \quad \mathbf{A1}$$

[5 marks]